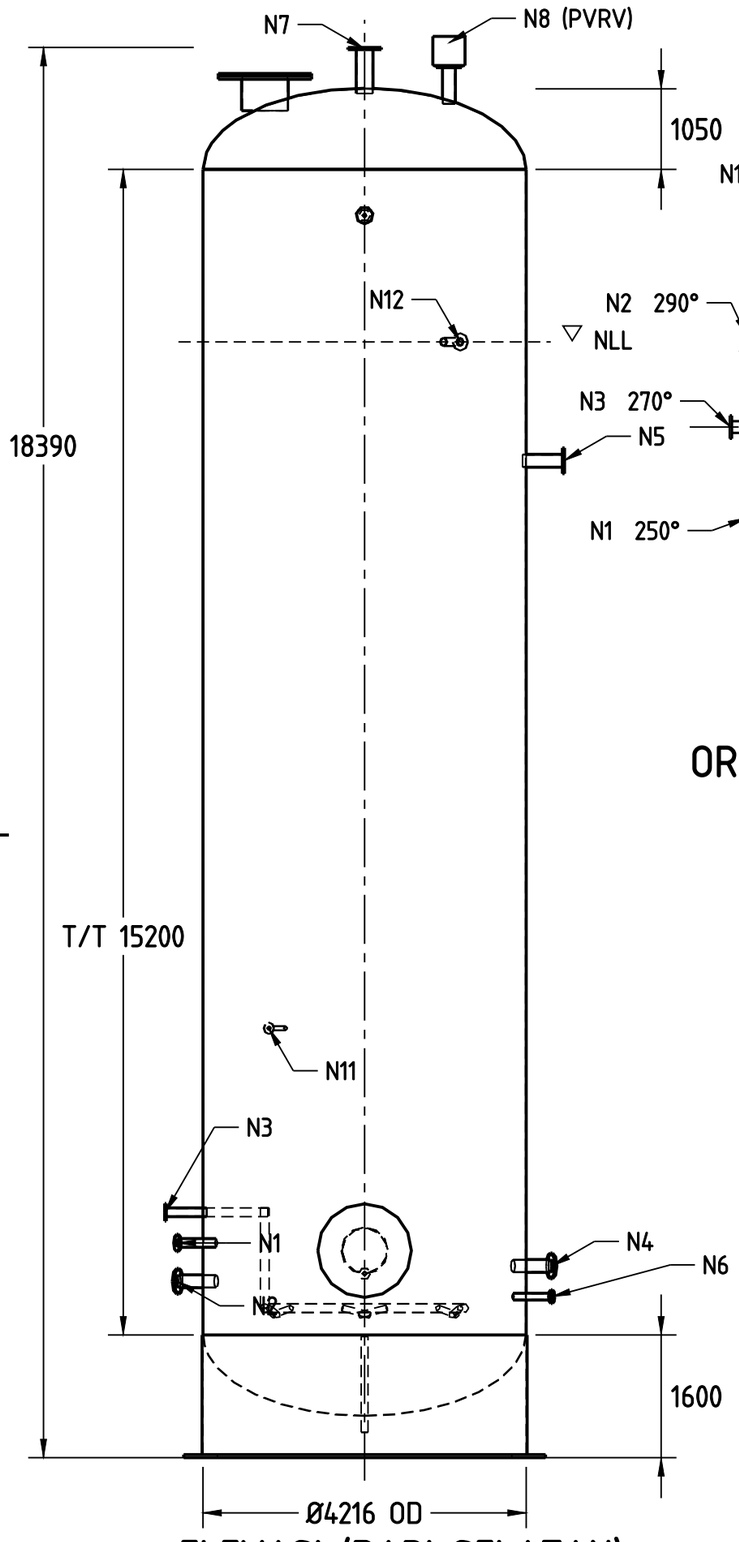
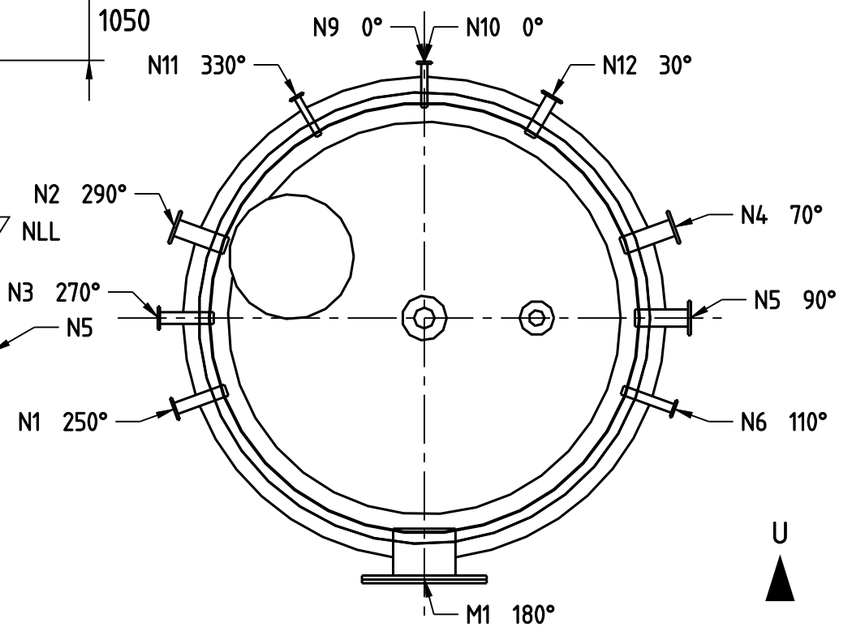


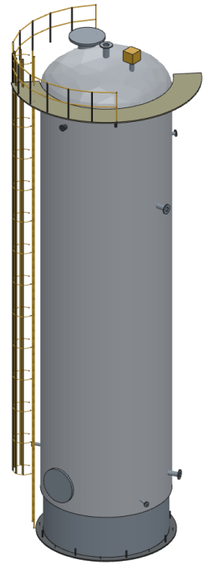

ELEVASI (DARI SELATAN)

SKALA 1:80




ORIENTASI NOSEL (TAMPAK ATAS)

SKALA 1:60



ISOMETRI (TANPA SKALA)

| JADWAL NOSEL | | | | | |
|---|---|---|---|---|---|
| Tag | Ukuran | Rating | Layanan | Elev. | Orient. |
| N1 | DN100 | PN10 RF | Umpan reaktor (dari P-201) | 2800 | 250° |
| N2 | DN150 | PN10 RF | Isap jet mixing (ke P-202) | 2300 | 290° |
| N3 | DN100 | PN10 RF | Balik jet mixing (header nosel jet) | 3200 | 270° |
| N4 | DN150 | PN10 RF | Lumpur ke membran (ke P-301) | 2500 | 70° |
| N5 | DN150 | PN10 RF | Retentat balik (dari MB-301) | 13000 | 90° |
| N6 | DN80 | PN10 RF | Lumpur buangan (ke P-303) | 2100 | 110° |
| N7 | DN200 | PN10 RF | Keluaran biogas | puncak | – |
| N8 | DN150 | PN10 RF | PVRV / relief tekanan-vakum | kepala atas | – |
| N9 | DN50 | PN10 RF | Pemancar level LT (atas / bawah) | 16200 | 0° |
| N10 | DN50 | PN10 RF | Pemancar level LT bawah | 2400 | 0° |
| N11 | DN50 | PN10 RF | Termometer TT / pH / NH4 | 5600 | 330° |
| N12 | DN100 | PN10 RF | Kaca intip muka cairan | 14550 | 30° |
| M1 | DN600 | PN10 RF | Lubang orang bawah | 2700 | 180° |
| M2 | DN600 | PN10 RF | Lubang orang atas | kepala atas | – |
| N13 | DN80 | PN10 RF | Pengurasan (kepala bawah) | 330 | – |

| DATA DESAIN | |
|---|---|
| Data desain | Nilai |
| Fungsi | Reaktor AnMBR termofilik (14 unit identik) |
| Kode desain | API 650 / ASME VIII-1 (acuan), bejana atmosferik |
| Tekanan desain | +50 / −5 mbar(g) + hidrostatik penuh |
| Suhu desain / operasi | 70 °C / 55 °C |
| Volume total / kerja | 230 m³ / 194 m³ |
| Bahan shell / kepala | SA-516-70, 8 mm, CA 3 mm, lapisan epoksi |
| Kepala | Elipsoidal 2:1, 8 mm |
| Insulasi | Mineral wool 100 mm + kladding Al 0,8 mm |
| Massa kosong / operasi | ± 20 t / ± 218 t |
| Transport | rebah di low-bed, 17.3 x 4.4 m |

CATATAN :

1. Elevasi diukur dari dasar cincin rok (EL 0).
2. Orientasi nosel: derajat dari utara, searah jarum jam.
3. Tangga, sangkar & platform atas: lihat isometri.
4. Bejana sama dipakai TK-101 / TK-301 (tanpa insulasi & internal).
5. Header jet internal (6 nosel) ditunjukkan garis tersembunyi.

| | | | | | | |
|---|---|---|---|---|---|---|
| University: UNIVERSITAS | Checked by : – | Design Lab. | Document type: Equipment Drawing | | | Rev: 0 |
| | Author : NAMA TIM | Reaktor AnMBR Standar R-201 s/d R-214 | Unit: mm | Scale: 1:80 / 1:60 | Creation date: 15/10/2026 | Size: A3 |
| | Eng. Approved by : – | | BG-V-201 | | | Sheet number: 2 / 8 |